

DVT UNIT-5

Syllabus:

Interaction Concepts:

Interaction Operators -Interaction Operands and Spaces - A Unified Framework

Interaction Techniques: Screen Space -Object-Space -Data Space -Attribute Space- Data Structure Space - Visualization Structure – Animating Transformations -Interaction Control.

Research Directions in visualizations: Steps in designing Visualizations – Problems in designing effective Visualizations – Issues of Data, Issues of Cognition, perception and Reasoning, Issues of system design, Evaluation, Hardware and Applications

Interaction concepts:

Interaction within the data and information visualization context is a mechanism for modifying what the users see and how they see it. Many classes of interaction techniques exist, including:

- navigation—user controls for altering the position of the camera and for scaling the view (what gets mapped to the screen) such as panning, rotating, and zooming.
- selection—user controls for identifying an object, a collection of objects, or regions of interest to be the subject of some operation, such as highlighting, deleting, and modifying.
- filtering—user controls for reducing the size of the data being mapped to the screen, either by eliminating records, dimensions, or both.
- reconfiguring—user controls for changing the way data is mapped to graphical entities or attributes, such as reordering the data or layouts, thereby providing a different way of viewing a data subset.
- encoding—user controls for changing the graphical attributes, such as point size or line color, to potentially reveal different features.
- connecting—user controls for linking different views or objects to show related items.
- abstracting/elaborating—user controls for modifying the level of detail.
- hybrid—user controls combining several of the above in one technique, for example, increasing the screen space assigned to one or more focus areas to enable users to see details, while showing the other areas of data in a smaller space, in a way that preserves context.

5.1 Interaction Operators:

In this section we describe in more detail a wide range of interaction operations commonly found in data and information visualization.

5.1.1 Navigation Operators:

Navigation (also sometimes referred to as exploration) is used to search for a subset of data to be viewed, the orientation of this view, and the level of detail (LOD). The subset in question may be one that is recognized by some visual pattern or one on which further or more detailed exploration is desired. In a typical three-dimensional space, this can be specified using a camera location, a viewing direction, the shape and size of the viewing frustum, and an LOD indicator. In multiresolution visualizations, LOD changes can correspond to drilling down or rolling up hierarchical representations of the data.

Navigation operators can work in absolute or relative coordinates within their particular spaces. Incremental navigation may have different granularities depending on whether the user wants a small or significant change. Navigation can be user-driven or automatic; a good example of automated exploration is the grand tour [22], where multidimensional data is explored by flying along a path that smoothly covers many or all possible orientations of the data space as shown in below Figure.

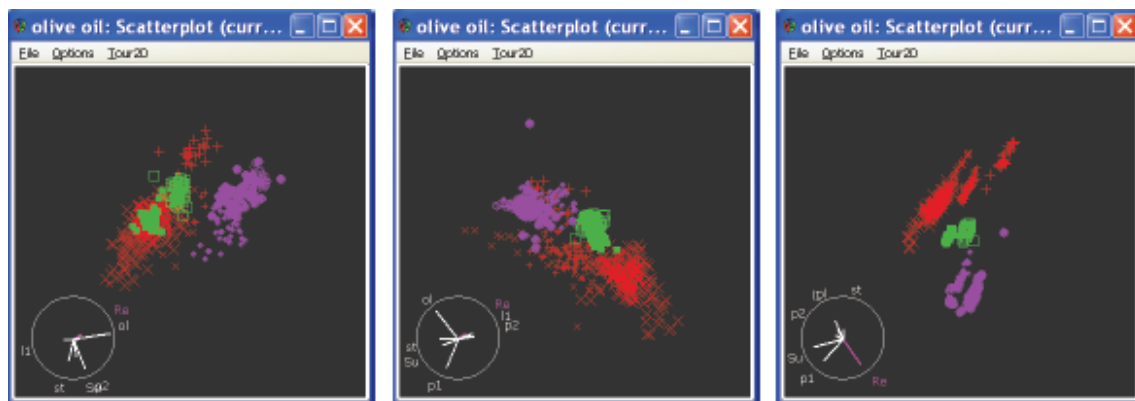


Figure .Views of the grand tour for the grades data set using GGobi

5.1.2 Selection Operators

In selection, the user isolates a subset of the display components, which will then be subjected to some other operation, such as highlighting, deleting, masking, or moving to the center of focus. Many variations on selection have been developed to date [461], and decisions need to be made on what the results should be for a sequence of selections. For example, should a new selection replace the previous selection or supplement the previous selection? The granularity of selection is also an issue. Clicking on an entity in the display might result in selection of the smallest addressable component (e.g., a vertex or edge) or might target a broader region around the specified location (e.g., a surface, region of the screen, or object).

Selection can be articulated in many different ways. The user may click on entities, paint over a selection of entities (e.g., holding the mouse button down while moving over the entities of interest), or perform selections in an indirect manner, where the system selects elements that match a user's input set of constraints. An example would be the selection of nodes in a graph that have a user-specified distance from a selected node.

5.1.3 Filtering Operators

Filtering, as the name implies, reduces the volume of data to be visualized by setting constraints specifying the data to be preserved or removed. A good example of such a filter is the dynamic query specification described by Shneiderman et al.. One- or two-handled sliders are used to specify a range of interest, and the visualization is immediately updated to reflect the changes made by the user. Range queries are just one form of filtering, however. One might also select items from a set or list to preserve or hide, such as the column hiding operation in Excel.

Figure below, shows the effects of filtering on a visualization to simplify the view and ease interpretation.

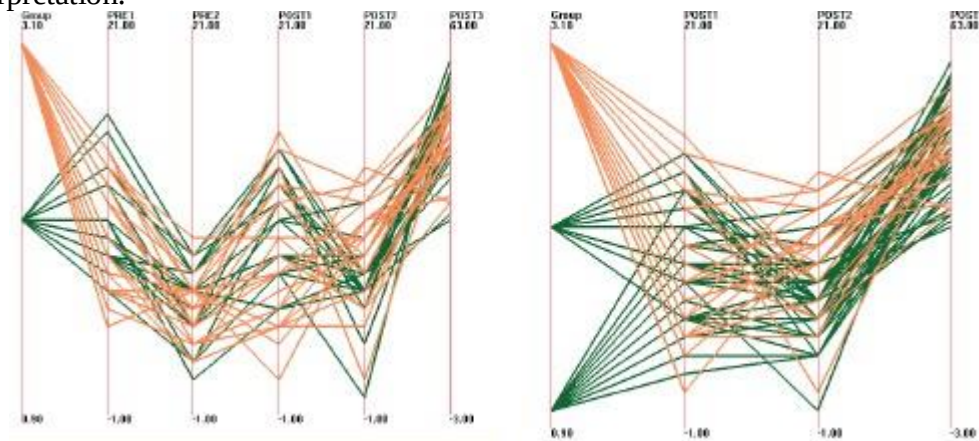


Figure . Filtering rows and columns of the grades data set using XmdvTool

5.1.4 Reconfiguring Operators

Reconfiguring the data within a particular visualization can often be used to expose features or cope with complexity or scale. By reorganizing the data, say by filtering some dimensions and reordering those that remain, different views are provided to the user. For example, a powerful tool with tablebased visualizations is to sort the rows or columns of the data to highlight trends and correlations. Other types of reconfiguration might be to change the dimensions being used to control the x- and y-coordinates of a plotted marker, or even to derive positions based on transformations of the data. Popular instances of this include the use of principal component analysis (PCA) or multidimensional scaling (MDS) to lay out data points so that relationships among all the dimensions are best conveyed in the 2D layout.

5.1.5 Encoding Operators

Any given data set can be used to generate countless different visualizations. Recoding can provide the user a library of possible different types of visualization; features of the data that are difficult or impossible to see with one such mapping might become quite apparent in another. For example, a scatterplot with one axis representing years may have many points that overlap, whereas a parallel coordinate display would represent these uniquely. Many visualization tools today support multiple types of visualization, because no single visualization is effective for all tasks that need to be carried out by the user. Each visualization is most suitable for a subset of data types, data set characteristics, and user tasks. While some work has been done to identify or select the best visualization, it is apparent that such guidelines are at best suggestions, and the analyst is most likely to benefit from examining their data using a number of different mappings and views. This is the essence of interactive visualization.

5.1.6 Connection Operators

A frequent use for selection operations is to link the selected data in one view to the corresponding data in other views. While other forms of connection between sub windows of an application exist, such as when opening a new data file, linked selection is probably the most common form of communication between windows found in modern visualization tools. Its popularity stems in large part from the fact that each view of one's data can reveal

interesting features, and that by highlighting such a feature in one view, it is possible to build a more complete mental model of the feature by seeing how it appears in other views. This can also help reveal relationships between this feature and others in the data set. For example, when examining multivariate spatial data, it is often useful to jump between the spatially referenced view and the dependent variable view, which often does not preserve the spatial attributes.

5.1.7 Abstraction/Elaboration Operators

In dense data and information displays, it is often desirable to focus in on a subset of the data to acquire details (elaboration) while reducing the level of detail (abstraction) on other parts of the data set. One of the most popular techniques of this type is using distortion operators. While some researchers classify distortion as a visualization technique, it is actually a transformation that can be applied to any type of visualization. Like panning and zooming, distortion is useful for interactive exploration.

5.2 Interaction Operands and Spaces:

Parameters of the interaction operators described in the previous section are discussed in more detail later in the chapter. First, however, we present our categorization of the interaction operands, as this will help clarify the role these parameters take in the interaction process and their semantics within the different spaces.

An interaction operand is the section of space upon which an interactive operator is applied. To determine the result of an interactive operation, one needs to know within what space the interaction is to take place. In other words, when a user clicks on a location or set of locations on the screen what entities does he or she wish to indicate?

5.2.1 Screen Space (Pixels)

Screen space consists of the pixels of the display. Navigation in screen space typically consists of actions such as panning, zooming, and rotation. Note that in each case, no new data is used; the process consists of pixel-level operations such as transformation, sampling, and replication.

Operations such as transformation, sampling, and replication. Pixel-based selection means that at the end of the operation, each pixel will be classified as either selected or unselected. As previously mentioned, the selection can be performed on individual pixels, rectangles or circles of pixels, or on arbitrarily shaped regions that the user specifies. Selection areas may also be contiguous or non-contiguous. Screen space filtering consists of cropping or masking a region of pixels.

Distortion in screen space involves transforming pixel positions to emphasize certain areas without hiding others. The transformation must preserve order and continuity. Magnification at any point is derived from this transformation, allowing control over the distortion effect. Common techniques include fisheye lens and rubber sheet methods, as shown in Figure below.

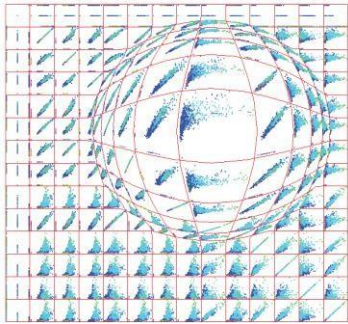


Figure: In screen space techniques, pixel regions are enlarged or reduced to provide selective detail. In this scatterplot matrix display, a center of focus has been selected and magnified using a confocal lens technique.

5.2.2 Data Value Space (Multivariate Data Values)

Operations on data space are applied directly to the data, rather than to the screen. Navigating in data value space involves using the data values as a mechanism for view specification. The analogous operations for panning and zooming would be to change the data values being displayed; panning would shift the start of the value range to be shown, while zooming would decrease the size of this range.

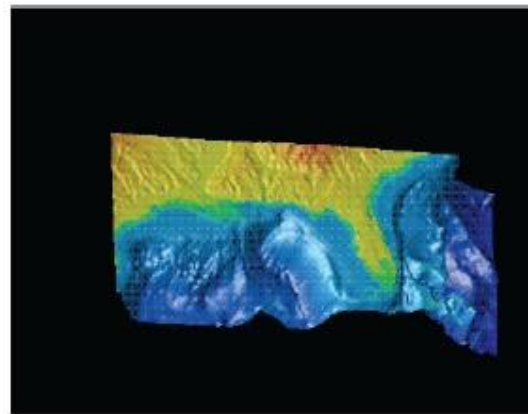
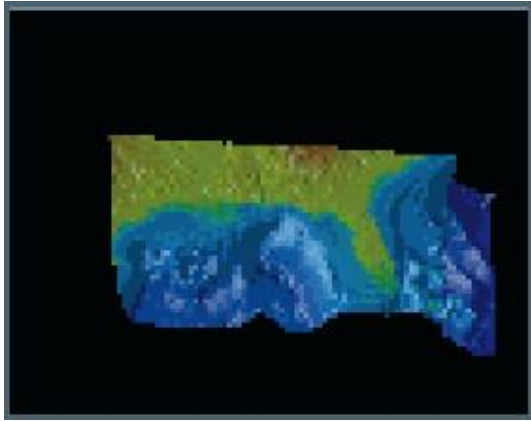
Data value space selection is similar to a database query in that the user specifies a range of data values for one or more data dimensions.

Filtering in data value space involves reducing large datasets to a specific region for easier visualization. For spatial data, this means clipping areas outside the view, while for nonspatial data, it involves removing certain records or dimensions. Sampling can provide a representative subset to avoid clutter, and dimension filtering helps focus on related or similar data characteristics.

5.2.3 Data Structure Space (Components of Data Organization)

Data can be structured in a number of ways, such as lists, tables, grids, hierarchies, and graphs. For each structure, one can develop interaction mechanisms to indicate what portions of the structure will be manipulated, and how this manipulation will be manifested. Navigation in data structure space involves moving the view specification along the structure, as in showing sequential groups of records, or moving down or up a hierarchical structure (as in drill-down and roll-up operations).

For example, Figure below shows the difference between a screen space zoom (involving pixel replication) and a data structure space zoom (involving retrieval of more detailed data). A technique presented by Resnick selects subsets of data to be visualized by specifying a focus, extents, and density in a regular grid structure, where the density can be a function of distance from the focus.



In screen space zooming (left), pixels are replicated to provide selective size, while in data space zooming (right), the data itself can be resampled at the appropriate resolution.

Filtering in data structure space reduces displayed information by focusing on specific parts of the data. In time-series visualizations, this means selecting a time range of interest. In graph visualizations, it involves showing only nodes and links within a certain distance from a focus point. In hierarchical data, users can filter by hierarchy levels to simplify the view.

Filtering in data structure space focuses on specific data parts—by time range, distance in graphs, or hierarchy level—to simplify visualization.

5.2.4 Attribute Space (Components of Graphical Entities)

In attribute space, operators are focused on one or more of the attributes associated with the graphical entity being used to convey information. Such attributes could include color, size, shape, or any other of the eight visual variables. Navigation in attribute space is similar to that in data value space; panning involves shifting the range of the values of interest, while zooming can be accomplished by either scaling the attributes or enlarging the range of values of interest.

As in data value-driven selection, attribute-space selection requires the user to indicate the subrange of a given attribute of interest. For example, given a visual depiction of a color map, a user can select one or more entries to highlight. Similarly, if data records have attributes such as quality or uncertainty, a visual representation of these attributes, accompanied by suitable interaction techniques, can allow users to filter or emphasize data according to the attributes. Remapping is often done in attribute space, either via selecting different ranges of an attribute to be used in the data to graphic mapping, or by choosing a different attribute to be controlled by the data. Attribute-based distortion modifies one or more attributes of the graphical objects used to depict the data.

5.2.5 Object Space (3D Surfaces)

In these displays, the data is mapped to a geometric object, and this object (or its projection) can undergo interactions and transformations. Navigation in object space often consists of moving around objects and observing the surfaces on which the data are mapped. The system should support global

views of the object space as well as close-up views. The latter may be constrained to enable the user to find good views more quickly. Selection involves clicking anywhere on the object(s) of interest, or indicating target objects from a list.

A typical example of remapping in object space would be changing the object upon which the data is being mapped, such as switching the mapping of geographical data between a plane and a sphere.

5.2.6 Visualization Structure Space

A visualization consists of a structure that is relatively independent of the values, attributes, and structure of data. For example, the grid within which a scatterplot matrix is drawn and the axes displayed in many types of visualizations are each components of the visualization structure, and can be the focus of interactions. Visualization dashboards, consisting of a set of tightly or loosely connected visualizations juxtaposed on the screen, have become increasingly popular for monitoring and analyzing complex data environments, such as for business analytics and command-and-control centers.

Examples of navigation in visualization structure space might include moving through pages in a spreadsheet-style visualization tool or zooming in on an individual plot in a scatterplot matrix.

5.3 A Unified Framework

For each interaction operator to be applied to a specified space/operand, several parameters are required. Some of these may be constants for a given system. The parameters are described below.

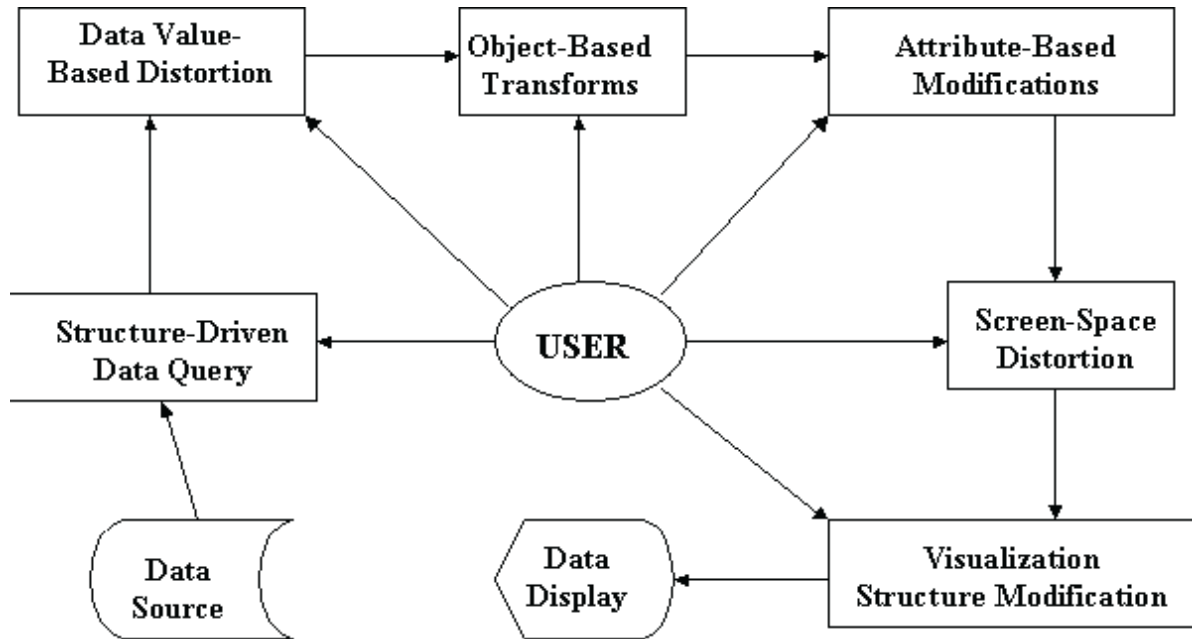
Focus. The location within the space at the center of the area of user interest. There may be multiple simultaneous foci, though for navigation this usually requires multiple display windows.

Extents. The range within the space (can be multidimensional) defining the boundaries of the interaction. The metric used for specifying the range is specific to the space; in screen space this would be in pixels, while in structure-space this might be the number of rows in a table or links in a graph.

Transformation. The function applied to the entities within the extents, generally a function of distance or offset from the focus. The shape of this transformation might also depend on the type of information being affected. For example, text distortion is more likely to have a flat peak to the transformation function. Another component of the transformation is the degree or scale factor for the transformation, thus allowing varying amounts of the specified action.

Blender. How to handle parts of space touched by more than one interaction. For selection, this operation may include performing logical operations on overlapping entities. For distortion, Keahey and Robertson identify several approaches, including weighted average, maximal value, and composition]. Each has advantages in terms of smoothness and ease of interpretation.

Figure below, illustrates a pipeline for the generalized distortion process, where users can control various operator parameters. While no system supports all stages, most visualization tools allow interactive control over some parameters. The order of operations can vary, though screen space distortion is usually applied last.



The distortion pipeline. The user interactively controls each stage of the pipeline. Each distortion operation is optional

Interaction Techniques

In this chapter, we discuss algorithm and implementation details pertaining to the interaction concepts.

5.4 Screen Space

Screen-space distortion is a common tool for providing focus+context. We present one example of such a distortion: the fisheye lens [140]. The implementation of a fisheye lens is rather straightforward. One needs to specify a center point for the transformation (cx, cy) a lens radius rl, and a deflection amount d. We then transform the coordinates of our image into polar coordinates relative to the center point. The lens effect is simply a transformation on the radius portion of the coordinates. One popular transformation is

$$r_{\text{new}} = s \log(1 + d * r_{\text{old}}),$$

(12.1) The overall pseudocode for the algorithm is as

follows:

1. Clear the output image.
2. For each pixel in the input image,

- (a) Compute the corresponding polar coordinates.
- (b) If the radius is less than the lens radius,
 - i. Compute the new radius;
 - ii. Get the color at this location in the original image;
 - iii. Set the color of the pixel in the result image.
- (c) Else set result image pixel to that of original image.

Depending on the transformation used in screen space distortion, it is possible to either leave gaps or to cause pixels to overlap on the output image. While the overlaps do not cause much concern (in some implementations, these are averaged), the gaps need to be resolved via interpolation.

Different functions give different shapes to the lens; for text visualization, it is common to use a piecewise linear function with a flat top to make reading easier.

5.5 Object Space (3D Surfaces)

One of the methods for navigating large document and data visualizations is via a perspective wall [283], which shows one panel of the view on a surface orthogonal to the viewing direction, with the rest of the panels oriented so they fade into the distance via perspective foreshortening. A simplistic version of the perspective wall can be created by noting that the front wall implements a horizontal scaling on a segment of the 2D image being mapped, while the adjacent segments are subjected to a horizontal and vertical scaling proportional to the distance to the edge of the front wall, along with a shearing transform. Thus if the left, center, and right sections of the original image to be plotted are bounded by $(x_0, x_{\text{left}}, x_{\text{right}}, x_1)$ and the left, center, and right panels of the result image are $(X_0, X_{\text{left}}, X_{\text{right}}, X_1)$, the transformation is as follows:

- for x

Given the perspective wall, the user could interact with it by sequential page movements (forward and backward), either one at a time or in a scanning or page-flipping manner. Indexes could also be used to jump to sections of interest, perhaps implemented as a tab sticking out of the top of the page at the start of each section. It might also be useful to have more than one page be in the direct focus, thus trading off some detail for a larger undistorted window on the data.

5.6 Data Space (Multivariate Data Values)

Data space transformations are actually quite common in data analysis and visualization. Some typical functions include:

- scaling and translating to fit a particular range of values;
- exponential or log scaling to spread or compress the distribution of the data;
- sinusoidal functions to help study cyclic behavior;
- absolute value transformation to focus on magnitudes of change;
- negating the values so that low values become high, and vice versa

A key requirement for these and other types of transformations is that the user be made aware that a transformation on the data has occurred. It is all too common that a data set is modified without the viewer being informed of this modification. Clear labeling of axes is essential; augmenting the visualization with a note describing any data transforms that have been applied reduces the chances of misinterpretation. Another key requirement is to translate and scale the resulting data values to fall within an acceptable range for the graphical entities and attributes being rendered. Failure to

incorporate this transformation can result in objects being mapped off the display, color wrap-around, negative values for graphical attributes, and other undesirable artifacts.

5.7 Attribute Space (Properties of Graphical Entities)

A great many interactions on attributes have direct equivalents with interactions in data space. Examples include global or local scaling, boundary enhancement, and color equalization. However, in some situations it makes more intuitive sense to consider the interaction to be simply on the graphical representation of the data, rather than the data itself. Thus, for example, when enlarging a data glyph of interest, the intuition is that you are not changing the data, just modifying the attributes of the glyph. There is also some overlap between the actions in attribute space and those in screen space, as the previous example hints at. If the glyph is not overlapping other glyphs, the enlarging can be based on pixel operations, rather than scaling the drawing primitives. In attribute space, however, you are more likely to avoid blocky artifacts that are typical of pixel-oriented zooming operations.

Perhaps the widest range of attribute space interactions involves modification of the color and opacity attributes. Techniques such as contrast enhancement, color histogram equalization, and others have been designed to make better use of the color space and make features of the data more readily perceived. For example, via controls for contrast and brightness, we can emphasize certain regions of the data to attract the attention of the analyst and make feature detection, classification, or measurement easier.

5.8 Data Structure Space (Components of Data Organization)

Most of the common structures for data (i.e., ordering, grids, grouping, hierarchies, and networks) have logical operations that users will want to interactively perform. In terms of the implementation, the main decisions to be made are the degree to which the operation can or should be automated, and whether the interactions will be specified directly on the visualization or in a separate dialog box. For automatic techniques, one must evaluate the trade offs between thorough, time-consuming techniques versus quick, yet suboptimal methods. In this section we will examine some of the design decisions that need to be made in providing these sorts of interactive operations.

Consider dimension ordering in multivariate data visualization. Fully manual techniques might involve manipulating text entries in a list, either via shifting operations (move up, move down) or drag-and-drop methods. In some visualizations, such as parallel coordinates or scatterplot matrices, direct manipulation of the axes may be possible. In the case of the scatterplot matrix, the movement of one element of the matrix would entail changes to entire rows and columns, in order to preserve the symmetry along the diagonal plots.

Another measure of goodness could involve the ease of interpretation. Different dimension orders can result in displays with more or less visual clutter or structure.

5.9 Visualization Structure Space (Components of the Data Visualization)

Several of the techniques described earlier can be readily adapted to work in visualization structure space. For example, the fisheye lens technique in screen space could just as easily be used in any of the visualization structures with sequences or grids of components; we could use the same

distortion function to adjust the spacing between axes in parallel coordinates or the size of the grid cells in scatterplot matrices or TableLens. As another example, we might adapt data clustering, filtering, or optimization actions (both manual and automated) to select and organize sets of visualizations on the screen, such as the results of performing volume visualization with hundreds or thousands of different parameter sets. Even drill-down, roll-up operations can be effectively utilized in multi-component visualizations that are hierarchically structured. One key is to make sure the user is aware at all times of the operations that can be done on the structure of the visualization, using consistent icons, labels, and visual representations that the user can quickly learn to use.

5.10 Animating Transformations

Virtually all user interactions within visualization systems will result in a change in the image being viewed. Some of these changes will be quite dramatic, such as when opening a new data set. Others may keep some aspects of the view the same, while changing other aspects. In situations where the user needs to maintain context while having his or her attention drawn toward the change, it is best to provide smooth transitions between the initial and final visualizations. Thus, for example, when rotating a three-dimensional object or data set, smoothly changing the orientation is generally better than simply jumping to the final orientation. In some cases, this might involve simply a linear interpolation between the starting and ending configurations, perhaps with the number of intermediate views dependent on the magnitude of the change. In some cases, however, linear interpolation does not result in a constant rate of change (such as moving a camera along a curved path). In addition, in most cases, a more appealing result can be obtained by performing smooth acceleration and deceleration of the change, rather than a constant velocity change. In this section we will discuss the algorithms necessary to achieve this control of changes. Students interested in a detailed exposure to these concepts will find them in textbooks on computer animation.

A commonly used curve for controlling animation involves an **ease-in, ease-out** behavior, where one accelerates from a velocity of 0 to a cruising speed and then decelerates to a speed of 0 when approaching the destination. Segments of a sine wave, with a straight line between them, can be used to approximate this behavior. The key is to maintain a smooth curve. Examples are shown in **Figure below**.

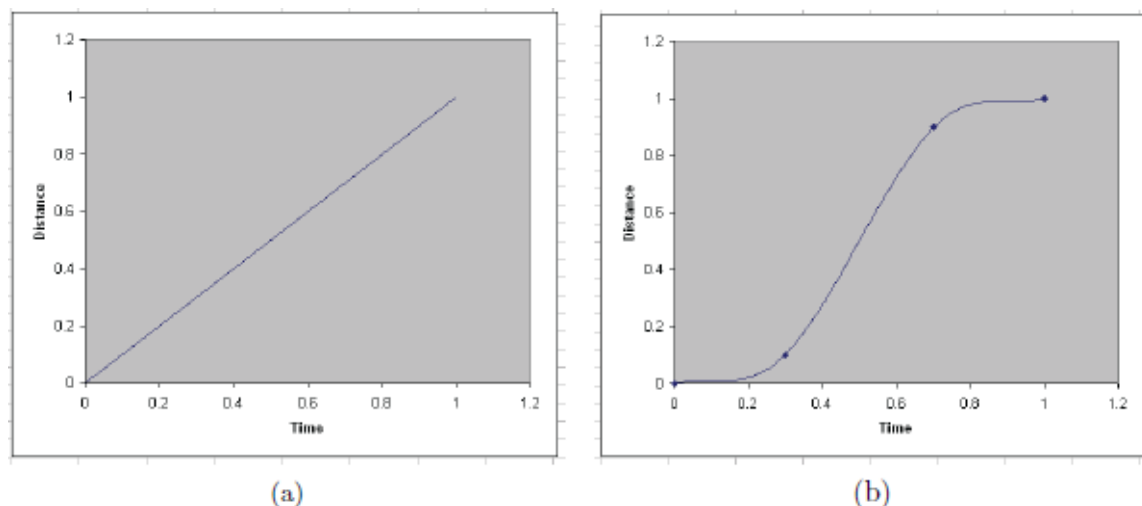


Figure . Examples of position curves:
(a) shows constant velocity;
(b) shows ease-in, easeout behavior.

5.11 Interaction Control

At every stage of the visualization pipeline, users need clear and intuitive controls to manage how they interact with data — including the type, location, and intensity of interactions. These controls must match the precision and scale of the space being used. The text outlines common types of controls and possible implementation methods, while noting that many other interface design strategies exist, as discussed in human-computer interaction research.

Focus Selection:

Selection is most readily accomplished via direct manipulation tools, e.g., using a mouse or other selection device to indicate the focus location. In screen and object space, this can be easily accomplished via normal selection operations. In attribute and structure space, a graphical depiction of the structure or the attribute range is first required—such as a display of a tree or table, or a curve representing the range of colors in the color map.

Extent Selection:

Specifying the extents for an interaction generally depends on the type of interaction and the space in which it is applied. It can be done either through direct manipulation or separate interface tools. The extent may be defined by a single value (e.g., a radius or maximum number of items), vector of values or a set of constraints. In many systems, the extents are often hard-coded to reduce the effort in performing the operation.

Interaction Type Selection:

Given the many possible types of interaction and the variety of spaces in which they may be applied, a suitable interface for this task would consist of a pair of menus—one to select the space and the other to specify the general class of the interaction. Icons or buttons in the user interface could also be used, as the number of choices would normally be modest.

Interaction Level Selection:

The degree of interaction is an important control parameter that can be defined by a single value (e.g., the magnitude of scaling at the focal point). A slider or dial is sufficient for this purpose, along with a button to reset the operation to its minimum level. A direct manipulation alternative could associate upward mouse movements with an increased interaction level, possibly combined with horizontal movements to adjust the extents.

Blender Type Selection:

If more than one interaction can be simultaneously viewed and manipulated, there must be some mechanism for selecting a strategy for mixing regions of space affected by more than one interaction. As with interaction-type selection, this is best accomplished via a menu of options.

Available options might be dependent on both the space in which the interaction is occurring and the type of interaction being used. As interactions in different spaces are applied at different points in the pipeline, it is necessary to consider methods for controlling the combination of interactions involving two or more spaces.

An important feature that should be present in all operations is the animation of interpolated values of the interaction parameters as they are changed. This has been shown to be extremely effective in many implementations of operators for helping users to both preserve context and to

obtain a better understanding of the effects of the operation on the data. Rapid changes can lead to confusion and a loss of orientation, especially when interactively exploring large data or information repositories. **Related to this**, users should have some control over the rate of this animation (the number of frames or steps in the interpolation).

5.12 Designing Effective visualizations:

The goal of this chapter is to provide some guidelines for designing successful visualizations. A successful visualization is one that efficiently and accurately conveys the desired information to the targeted audience, while bearing in mind the task or purpose of the visualization (exploration, confirmation, presentation). For any particular set of data there is a myriad of possible methods for mapping data components to graphical entities and attributes. Similarly, there exists a wide range of interactive tools that the user may be provided. Selecting the most effective combinations of techniques is by no means a straightforward process.

A visualization may be ineffective for a number of reasons. It might be too confusing or complex to be interpreted by the intended audience, or some of the data may have been distorted, occluded, or lost during the mapping process. Other signs of deficient visualizations are the lack of support for view modification or color map control. Even aesthetics can influence the success of a visualization; a visually unappealing presentation can affect an audience's willingness to look at the images. In each of the above cases, some component of the visualization is interfering with the delivery of information to the user.

This chapter first presents design considerations for the components that the authors feel are necessary for a good visualization. Following this, we explore some of the common problems found in visualizations and propose some techniques for avoiding these problems.

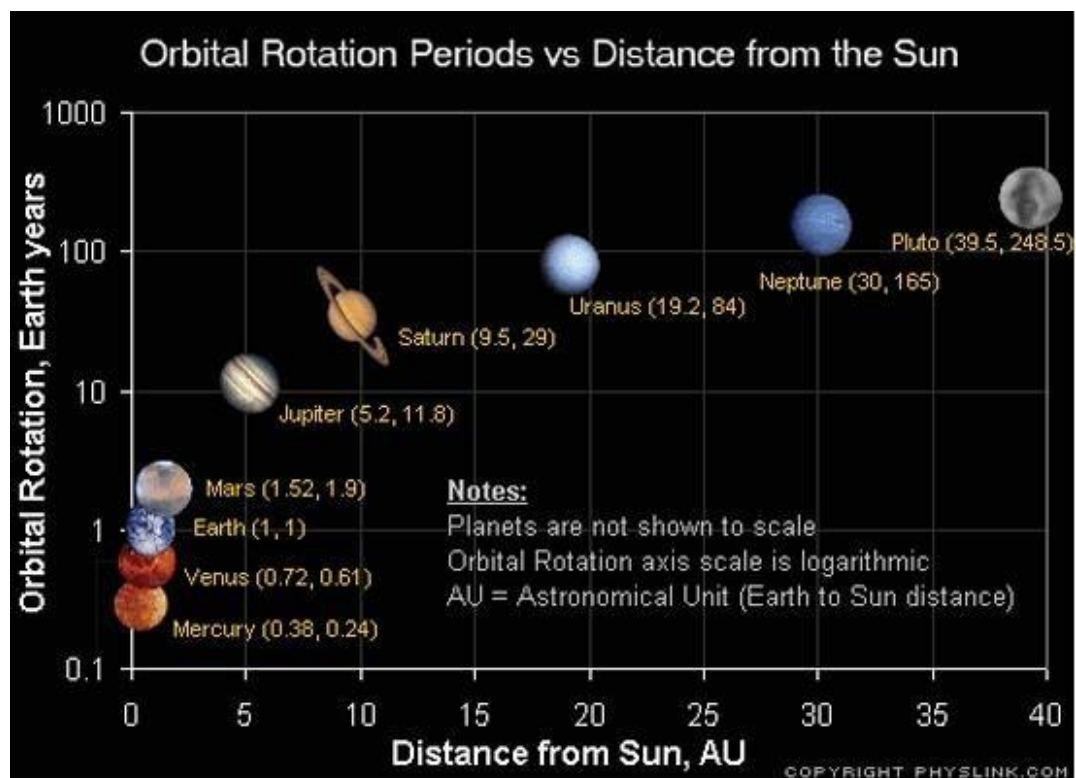
Steps in Designing Visualizations

Creating a visualization involves deciding how to map the data fields to graphical attributes, selecting and implementing methods for modifying views, and choosing how much data to visualize. Additional information regarding the data being shown (e.g., labels) and the mapping (e.g., a color key) are also essential to facilitate interpretation, and must be integrated into the visualization. The final, less tangible, consideration is the overall aesthetics of the resulting display. In this section we present, for each of these design stages, some issues that should be addressed by the visualization designer.

(a) Intuitive Mappings from Data to Visualization

To create the most effective visualization for a particular application, it is critical to consider the semantics of the data and the context of the typical user. By selecting data-to-graphics mappings that cater to the user's domain-specific mental model, the interpretation of the resulting image will be greatly facilitated. In addition, the more consistent the designer is in

predicting the user's expectations, the less chance there will be for misinterpretation. Intuitive mappings also lead to more rapid interpretation, as translation time is reduced. For example, in Figure , images of planets are used to plot the relationship between the distance from the planet to the sun and the duration of its orbit. Mapping spatial data attributes, such as longitude and latitude, to screen position is perhaps the most common and intuitive mapping found in visualizations. Some of the earliest visualizations took advantage of the ability of humans to correlate position on the drawing medium with position in the three-dimensional world. Likewise, with the advent of animation, it is obvious that displaying temporally related data sets via animation is reasonably intuitive, with the added advantage of allowing time to vary in both speed and direction,



Using intuitive scatterplot symbols to show the distance from planets to the sun versus the duration of a single orbit. (Image from <http://www.physlink.com>.)

One of the important considerations when selecting a mapping is the compatibility between the scale of the data field and that of the graphical entity or attribute. For ordered data attributes (e.g., age), it is not reasonable to select a graphical attribute that is not ordered, such as shape. Similarly, unordered data attributes (e.g., country of origin) should not be mapped to ordered attributes (e.g., length).

(b) Selecting and Modifying Views

Except for fairly simple data sets, one view is rarely sufficient to convey all the information contained in the data. The key to developing an effective visualization is to be able to anticipate the types of views and view modifications that will be of most use to the typical user, and then provide intuitive controls for setting and customizing the views. Useful views,

as mentioned earlier, depend heavily on the type of data being presented, and the task associated with the visualization. Each supported view should be clearly labeled, and selecting a new view should require minimal actions on the user's part.

View modifications fall into a number of categories, and their inclusion as part of the functionality should be considered based on user priorities.

Scrolling and zooming operations are needed if the entire data set cannot be presented at the resolution desired by the user

- Color map control is almost always desirable, minimally supporting a set of different palettes, and preferably offering the user control of either individual colors or the complete palette.
- Mapping control allows users to switch between different ways of visualizing the same data. Features of the data that are hidden in one mapping may stand out in others.
- Scale control permits the user to modify the range and distribution of values for a particular data field prior to its mapping. Similarly, clipping and other forms of filtering allow the user to focus on data subsets.
- Level-of-detail controls provide the ability to eliminate or highlight detail, supporting views at different levels of abstraction. Depending on the task at hand, a user may need to repeatedly switch between several distinct levels .

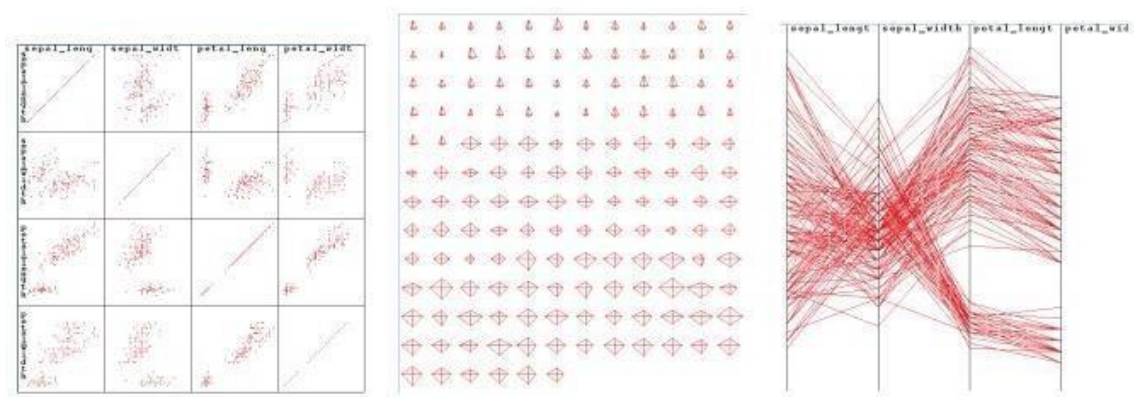


Figure 13.2. Three views of the Iris data set (scatterplot matrix, star glyphs, and parallel coordinates). (Image from XmdvTool.)

(c) Information Density—When Is It Too Much or Too Little?

One of the key decisions one makes when designing a visualization is determining how much information to display. This gives rise to two extreme situations. The first, which might be called “gratuitous graphics,” occurs when there is very little information to present. Many examples of graphics can be found that convey only two or three distinct values, such as the percentage of males and females within a particular sample (this actually can be communicated with one number). Others can be found that “pad” the number of pieces of information by deriving additional quantities, such as showing two numbers, their sum, and their difference. In cases such as these, it is often more effective to simply display the quantitative values as text. This requires much less screen real estate (which in many applications is quite valuable), while still getting the message across. It must be remembered that simply because one can create a visualization doesn’t imply that one must do so.

The other extreme, trying to convey too much information, is also a common problem. Excessive information content can lead to confusion, intimidation, and difficulties in interpretation on the part of the viewer. Important information contained within the data

can be lost or deemphasized on a cluttered display, and viewers may have a hard time determining where to focus their attention.

There are many effective solutions to the problem of excessive information content in a visualization. One method is to give the user the option of disabling or enabling different components of the display. In this manner, a user can decide which parts are most important to her, and can have the less important information displayed on demand.

Another solution is to use multiple screens, either as disjoint panes or with partial occlusion. This method makes better use of screen space, while making each of the individual pieces of data readily available.

Another common cause of cluttered displays is large or unevenly distributed data sets. As mentioned in the previous section, data sets may be filtered to remove uninteresting data points, allowing the user to concentrate only on the significant parts. Similarly, uneven distributions, which might lead to some parts of the screen being congested, while others are sparsely populated, can sometimes be rectified through scaling of one or more data dimensions.

(d) Keys, Labels, and Legends

A common problem with many visualizations is that insufficient information is provided to the user to allow unambiguous and accurate interpretation. This supporting information should begin with a detailed caption indicating the particular data fields being displayed, and the mappings that were used.

Additionally, grid or tick marks should be displayed to convey the ranges and values of interest for numeric fields when absolute judgments are important, and all axes should be labeled with appropriate units. If symbols are being used, a key must be provided, either along the border of the display or within a separate widget. Finally, if color has a significance, sufficient information must be available to allow easy interpretation (e.g., via a labeled color bar).

The use of grid and tick marks can be both a boon and a curse to the visualization. Poor choices of the types of markings and the density used. The actual positions of the markings can also have a bearing on how readily the data is interpreted. Based on the semantics of the data, certain gaps between markings may make more sense to the user than others. Unfortunately, the default values used by some visualization tools may make correct interpretation difficult.

One final rule of thumb pertains to the use of multiple frames or windows. It is important to follow a consistent labeling and gridding scheme. Changing the position of labels and keys or the range of values shown (field) can cause confusion and increase the risk of misinterpretation. If range changes are necessary (e.g., for views that differ in level of detail), the label, as well as the grid markings, should convey the change. Similarly, if different color mappings are necessary, the visualizations must clearly convey this information.

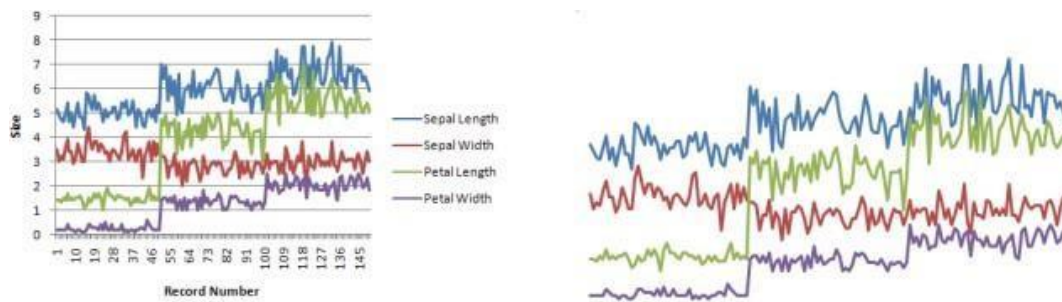


Figure 13.4. A complex visualization with and without captions/ticks/legends.

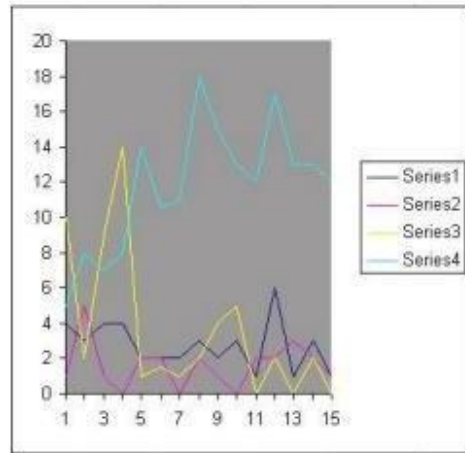
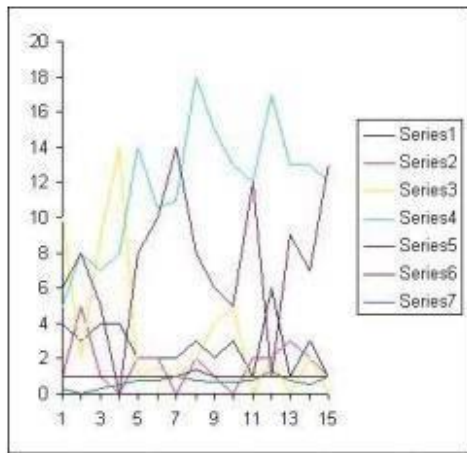
(e) Using Color with Care

One of the most frequently misused parameters in visualization design is that of color. Selecting the wrong color map or attempting to convey too much quantitative information through color can lead to ineffective or misleading visualizations. Also, since color perception is context-dependent (a particular color will appear quite different, depending on adjacent colors), the characteristics of the data itself can influence how the colors are perceived. Finally, it must be remembered that many people are color blind or color confused; it has been determined that as many as ten percent of all males have some form of color deficiency.

The following guidelines can assist in the effective use of color in visualization.

1. If the visualization task involves absolute judgment, keep the number of distinct numeric levels low
2. Use redundant mappings if possible, e.g., map a particular field to both color and size , to improve the chances of the data being communicated accurately.
3. In creating a color map for conveying numeric information, make sure that both hue and lightness are changed for each entry
4. Include a labeled color key to help users interpret the colors .
5. When possible, use semantically resonant colors in the visualization , these will be easier for users to learn and remember.

Color can add significant visual appeal to a visualization, but can also significantly decrease the effectiveness of the communication process. Some interface designers advocate an initial design process that only involves the use of grayscales. Once this design has been refined and tested, the addition of color can usually be done in a more effective manner.



Too many colors versus a moderate number of colors.

(f) The Importance of Aesthetics

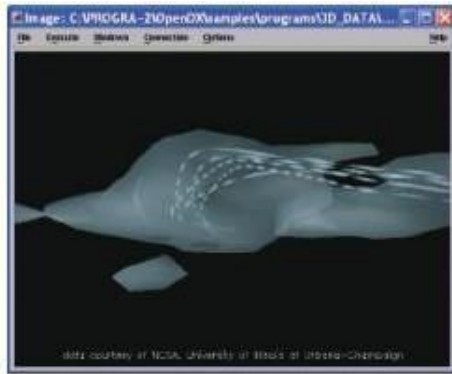
Once we have ensured that our designed visualization conveys the desired information to the user (function), the final step is to assess the aesthetics (form) of the results. The best visualizations are both informative and pleasing to the eye. In contrast, a visualization that is visually unappealing can detract from the communication process. An aesthetically pleasing visualization invites the viewer to study it in depth.

There are many guidelines for attractive visualization design that can be drawn from the art and graphic design communities. These include:

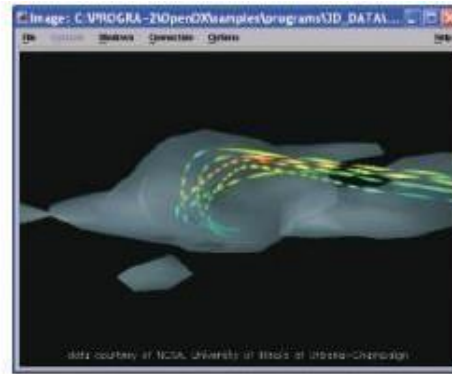
Focus: The viewer's attention should be directed toward the most important part of the visualization. If the key components are not sufficiently emphasized, viewers may lack clear cues for guiding their inspection.

Balance: The screen space should be used effectively, with the most important components placed near the center. Emphasis should not be given to any particular border.

Simplicity: Avoid cramming too much information into a single display and do not use graphical effects simply because they are available (for example, using 3D Phong-shaded histograms when a bar or line chart would convey the same information). A useful approach after designing a visualization is to iteratively remove features and evaluate the loss of information conveyed. Features whose removal results in minimal information loss can likely be discarded.

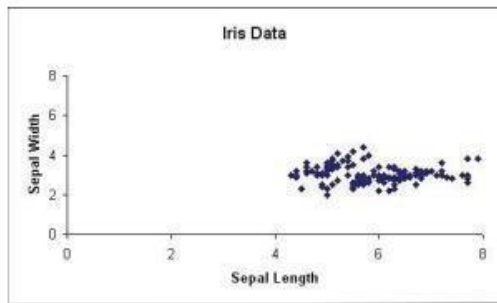


(a)

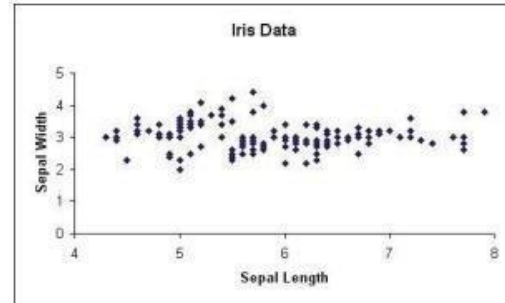


(b)

(a) Subdued streamlines vs. (b) highlighted streamlines from OpenDX.



(a)



(b)

Figure 13.12. (a) Everything to one side vs. (b) balanced between left and right.

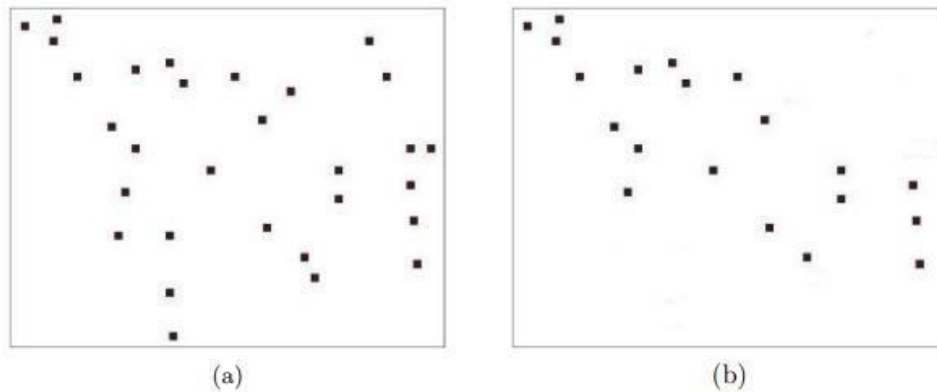
5.13 Problems in Designing Effective Visualizations:

(1) Misleading Visualizations: One of the foremost rules of visualization is that the image should be an accurate depiction of the data. However, throughout history, there are examples of how visualizations from distorted data have been used to sway opinions and lie to the audience. These so-called “viz lies” can be found everywhere, the most prestigious journals to company portfolios. In this, we identify some of the common strategies for creating misleading visualizations, not for the reader to practice, but to try to avoid!

Data scrubbing. Raw data can often be very rough in form, and the temptation when creating a visualization is to remove some of the roughness.

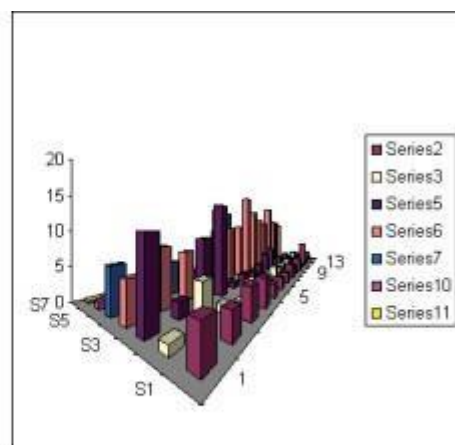
Unfortunately, sometimes the selection of which data to remove is biased to eliminate data that does not support a particular point that the author of the data is espousing (see Figure 13.15). Outlier removal is a common tactic in this situation. Unless there is reason to believe that the outliers resulted from flaws in the data acquisition process, they

should not be removed without informing the viewer and providing the option for the outliers to be displayed



15. The problem with data scrubbing: (a) raw data showing lack of correlation; (b) scrubbed data revealing false correlation.

Unbalanced scaling. Scaling is a powerful tool in visualization, since careful selection of scale factors can reveal patterns and structures not visible in unscaled views. However, scaling can be used to deceive the viewer into believing that a trend is stronger or weaker than supported by the data. This can lead to what Tufte refers to as the *lie factor* [424], which is the ratio between the raw data change and the change as depicted in the visualization.



1. Vis Lies: perspective distorts size in favor of closer objects.

Range distortion. As mentioned in an earlier section, viewers often have an expectation about the ranges for a particular data dimension; by setting this range to be significantly different from this expectation, the user may be deceived into misinterpretation. This is often done by moving an axis so it no longer corresponds with the expected “zero value”. Since relative judgment is such a strong component of our perceptual system, changing the baseline for the relations being portrayed could have a serious effect on how the image is interpreted. The designer may want to give the user the option of moving

this baseline to avoid wasting screen space, but it should be made clear what the baseline is, especially if it departs from the established norm.

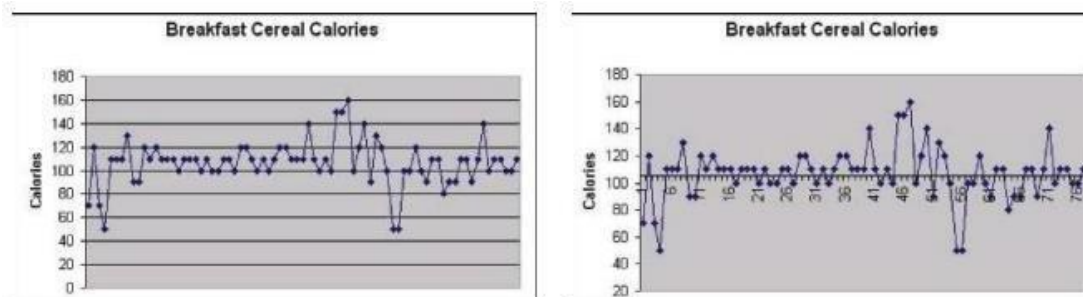


Figure 13.17. Plotting data with different baselines.

Abusing dimensionality. we noted that errors in interpretation rise with the power of the dimensionality being portrayed. Thus, our errors in judging volume are much worse than those for area, which in turn are worse than those for length. Therefore, mapping a scalar value to a graphical attribute such as volume can dramatically increase the likelihood of erroneous interpretation. As mentioned earlier, it is often the case that simpler is better.

1. Visual Nonsense:

Visualizations are designed to convey information, and it is essential that the information be meaningful. Visualizations are often created by combining data sets from different sources. However, it is easy to merge unrelated components into a single visualization and perceive what seems to be structure; for example, plotting stock market values against occurrences of sunspots. In such cases, coincidental relationships can be mistaken for causal ones.

When deciding what data to combine, it is important to ensure that there is some logical basis for the combination. One common problem in analytic pattern recognition or data mining processes is that irrelevant relationships are often discovered and reported, requiring a domain specialist to eliminate them. The visualization designer should aim to avoid creating such meaningless graphics before presenting them to users.

Another factor to consider is the compatibility between temporal and spatial ranges for the data being compared. For instance, one should not compare the sales of a particular product in one year for a specific region with the sales of the same product in a different region and year, unless hypothesizing a migration of interest in the product.

Compatibility in measurement units also needs attention when preparing a data set for visualization. For example, food products may be measured in terms of price per volume or price per weight. An effective visualization might normalize both to price per serving for meaningful comparison.

2. Losing Data in the Chart Junk:

Chart junk can be defined as any supplementary (nondata) graphics in a visualization that are not necessary for the accurate interpretation of the data. This additional information can lead not only to visualizations that appear overly complex, but also to occlusion and deemphasis of the actual data.

Deciding the amount of supplementary graphics to include in a visualization is sometimes a difficult process, since the designer might not know the needs of all potential users. However, because we are dealing with a dynamic, customizable medium (unlike Tufte's static charts), the option exists to allow users to adjust the types and density of this supporting information on the display.

In some visualization tasks, users can switch between qualitative overviews and quantitative analysis. In the former case, it is usually more important to give viewers a clear view of the data, while in the latter case, tools to help quantify the elements of the display are much more desirable.

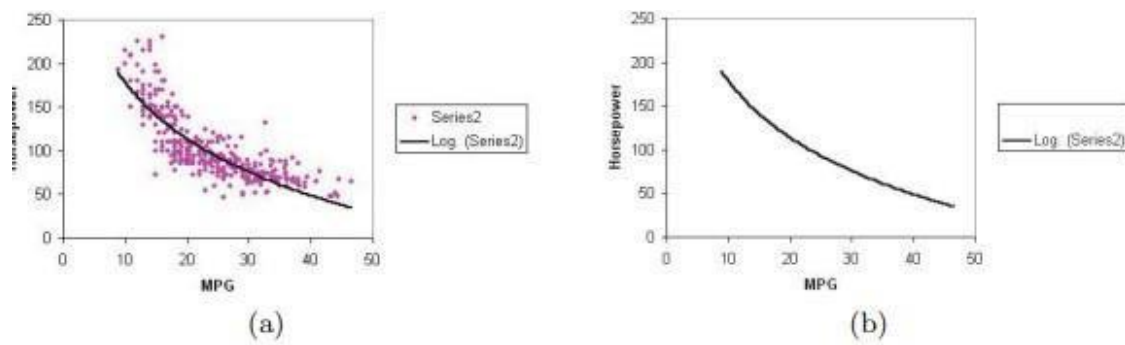
Thus, a good rule of thumb is to provide sufficient tools to support the user's quantitative needs, but with the option of disabling them or altering their degree of presence in the visualization.

3. Raw versus Derived Data

A common practice is to compute an analytic model of the data using curve or surface fitting to obtain a more visually appealing result. Again, this distorts the truth and may lead to false assumptions and conclusions on the part of the observer.

In some visualizations, it is common practice to discard all of the raw data and show only the smooth approximation derived from that data. This forces the viewer to trust that the approximation is an accurate portrayal of the data, which is often not the case when the designer blindly applies statistical fitting algorithms. It is best to show both the raw data and the fitted model, and to allow one or the other to be deemphasized or filtered out on demand.

Yet another form of cleaning the data is the process of resampling, where raw data positioned either on a sparse grid or randomly are used to create data that are denser or on a regularly spaced grid. This can result in a much richer visualization, approaching that of continuous sampling, but it again deceives the user into believing the data set is much larger than it actually is. The denser the resampling, the more likely the user will misinterpret the data, unless the phenomenon being observed has little variability.

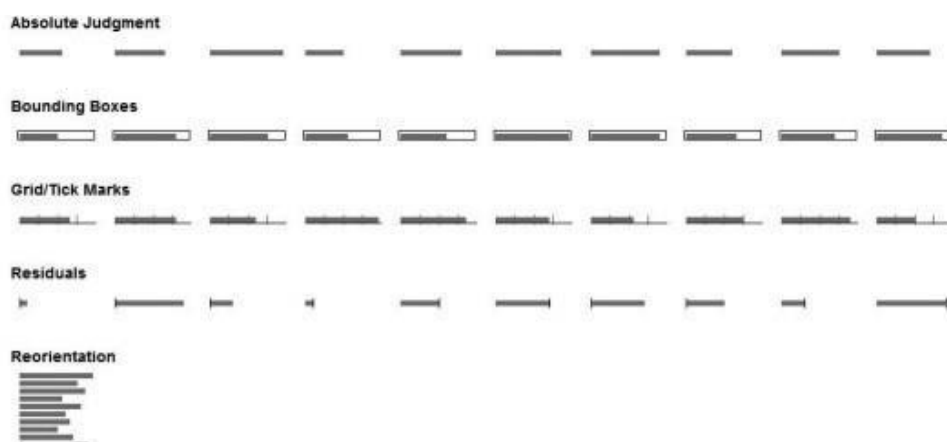


4. Absolute versus Relative Judgment

Humans have a fairly limited ability to make absolute judgments of visual stimuli. This implies that visualizations that depend too heavily on users performing accurate measurements of graphical attributes such as position, length, and color will lead to problems in interpretation.

One way to address this human limitation is to design visualizations that rely on relative rather than absolute judgment, or that use only a small number of distinct values for each graphical attribute conveying information.

Bounding boxes, grids, and tick marks are all effective tools for converting an absolute judgment task into one that depends more on relative judgment. By comparing the length or position of a graphical element against a quantified structure, users can more easily determine the approximate value relative to known levels. Using residuals (for example, subtracting values from their means) can also transform a measurement task into one of determining whether a value is above or below a particular level.



Some examples of absolute versus relative judgment. (Image courtesy of Michael Barry.)

5.14 Research Directions in Visualization

Visualization is now a sufficiently mature field, with many groups of researchers dedicating their efforts to contemplating and predicting its future directions.

1. Issues of Data

Many of the current and proposed future research activities are centered on expanding the characteristics of data that can be effectively visualized. Some of these are discussed here.

Scale.

Perhaps the most frequently addressed problem dealing with data is finding solutions for coping with ever-increasing sizes of data sets. For scientific domains, both acquired and simulated data have been growing in orders of magnitude, certainly outpacing Moore's Law. While only a few years ago researchers were focused on data sets of modest size (kilo bytes to a few megabytes), it is now common to be looking at problems involving gigabytes and terabytes of data. Solutions such as clustering, sampling, and abstraction are all being actively pursued in areas such as graph visualization, genetic sequence analysis, and large-scale flow simulations. Massive data, that which does not fit into memory, is now commonplace.

Static versus dynamic.

While most visualization techniques to date have been developed with the assumption that data is static (e.g., in files or databases), a growing interest is in the area of visual analysis of dynamic data. An increasing number of streaming data sources are being studied in the database and data mining communities, and efforts to perform visual analysis on this type of data are starting to emerge. The basic concept is that the data is continually arriving, and has to be analyzed in real time, both because of the urgency with which analysis must be performed, as well as the fact that the volume of data precludes its permanent storage. Analysts are interested not only in the values at a particular time period, but also in how the data is changing. Since the data is so large in many cases, the data cannot even be stored as it is streaming.

Spatial versus nonspatial data.

A growing number of application areas for visualization include both spatial and nonspatial data, including many scientific and engineering fields. To provide analysts with a powerful environment for studying this data, several recent efforts have focused on the integration of the spatial visualization techniques normally found in scientific visualization with the nonspatial techniques that are common in information visualization. The linkages between these distinct types of views is critical to successful analysis.

Nominal versus ordinal.

The graphical attributes to which we map data in our visualizations, such as position, size, and color, are primarily quantitative in nature, while it is quite common to have data that is not quantitative, such as the name of a gene or the address of an employee. If this nominal data is to be used in the visualization, a mapping is needed. However, it is also important to ensure that relationships derived from visual analysis are truly part of the data, and not an artifact of the mapping. Some solutions

include selecting color schemes and plot symbols that do not impose a perceptual ordering; others attempt to assign a numeric mapping of the nominal variables that preserves similarities and differences implied by the data. For example, in a data set containing statistics about cars, two cars with similar characteristics in their attributes might have similar numbers assigned to the name field.

Structured versus nonstructured.

Data can be classified based on the degree to which it follows a predictable structure. For example, tables of numbers are considered highly structured, while newspaper articles are regarded as unstructured. Between these extremes lies semi-structured data, such as an email message that contains both a structured component (sender, time, receiver) and an unstructured part (message body).

Key research problems include extracting useful information from the unstructured components of data sets and using the structured components to index and organize the data records. Text analysis and visualization have made great progress, but the research is still in its infancy.

Time.

Time is a special variable (and attribute of data). Time in dynamic data provides one view: a volume visualization over time deals with a physical representation, and a common interactive visualization uses time as a control. Spatio-temporal databases and queries and the visualization of results are becoming prominent as more data is being made publicly available. Time here can be handled just as in the volume visualization. But one can do more. For example, in a data set involving education and health indicator data, a useful question is how to identify similar patterns involving not just the data, but time as well. This begs a key research question of how to handle time as just another variable, and what are the ramifications

Variable quality.

While most visualization systems and techniques assume the data is complete and reliable, in fact most sources of data do not match these constraints. Often, there are missing fields in the data, often due to acquisition problems (e.g., defective sensors, incomplete forms). The quality of the data itself may also be problematic; out-of-date information can have low certainty associated with it, inaccurate sensors may produce values with significant variability, and manual entry of data can be error-prone. There is a great need for visualization technology that can cope with this data quality issue, this uncertainty, not only in its visualization, but also in keeping the analyst informed, so that the quality information can be incorporated into the decision-making process.

2. Issues of Cognition, Perception, and Reasoning

Many of the foundational concepts in data and information visualization have their roots in our understanding of human perception, particularly in aspects of selecting effective mappings of data to graphical attributes such as color and size. Much less work, however, has built on our knowledge of human cognition and reasoning processes, with the notable exception being the Gestalt laws to better understand how visual grouping is performed. Thus, a far-reaching research agenda for the visualization field must include the study of how humans do problem solving with support from interactive visual displays, and how we can leverage this knowledge to design even more effective and powerful visual tools.

Part of this effort will be directed at raising the level at which visual tasks are performed. For example, rather than using the visualization to identify a cluster or trend, it might instead be focused on building a mental model of the entire phenomenon being studied. While this task may be comprised of several more primitive subtasks, the visualization tool will need to support the combination and sequencing of these tasks to enable the analyst to cope with the scale and complexity of the data and information being gathered and generated. Tasks such as discovering associated patterns of change in the data will involve not only visualization of the data, but also how the data is changing and how those changes may be associated with other changes taking place.

These higher-level discoveries can then be used by the analyst to form, confirm, or refute hypotheses, expand or correct mental models, and provide confidence in decision-making processes.

Beyond decision making, we can also envision the expansion of visualization in the process of human learning. Educational opportunities for data and information visualization abound, but it is critical in the design of tools to support education that the theories of how humans learn be components of the design. People vary in the ways in which they learn most effectively: concrete versus abstract, sequential versus random. Different visual tools and mechanisms are likely to be needed to address these very different styles of learning.

In both problem solving and learning activities, we can also imagine using visualizations as a mechanism to expand and support the memory process, which is critical to both activities. Providing visual representations of intermediate results or supporting evidence, as well as structuring these artifacts on the screen, can help people remember facts and relationships in the same way that writing notes to oneself can relieve the human memory system of having to perform accurate recall. The extents to which visualization can be applied to this have yet to be extensively studied and exploited. Finally, and perhaps most importantly, we would like to measure cognitive primitives in the transfer from the data to the screen (easy) and from the screen to the human (hard).

3. Issues of System Design

One of the most crucial research challenges in developing visualization tools is determining how best to integrate computational analysis with interactive visual analysis. While many visualization systems support a modest number of computational tools, such as clustering, statistical modeling, and dimension reduction, and similarly many computational analysis systems support some amount of visualization, such as visualizing the analysis results, there have been no systems developed to date that provide a truly seamless integration of visual and computational techniques. Clearly, there are many tasks for which human perception and cognition are the most effective means to a solution; however, there are likely comparable numbers of tasks for which computational solutions are far superior. Indeed, in many situations, one can easily imagine using computational methods to guide the visual exploration, and vice versa. For example, one can use visual overviews of data to decide on appropriate parameters for filtering and clustering algorithms, after which the analyst could examine a resulting cluster to help select a computational model that best fits the characteristics of the data subset. Both visual and computational methods could then be used to ascertain the goodness of fit for the model. Many such scenarios can be easily envisioned. Another key problem.

Another key problem in visualization system design is the development of powerful new interaction paradigms to support the user's tasks. Many researchers believe that existing modes of interaction during visual analysis are ill-suited to the tasks being performed. This is likely due to the fact that while advances in hardware and visualization techniques have been moving by leaps and bounds, interaction methods have expanded at a much slower pace. While exciting and novel interaction tools can be seen in the immersive environment field, these and other types of interactions are either too lowlevel or are not designed with a particular high-level task in mind. What is needed is a fresh look at how users can interact with their data and information to perform specific tasks, such as determining the underlying phenomena or processes that are generating the data; understanding causal relationships; and designing, executing, and assessing "what if" scenarios. While existing low-level interactions such as point-and-click and drag-anddrop may be parts of the solution, there likely is much more needed to make visual analysis more efficient and effective,

Another issue is that most visualization systems require an expert user. Visualization in the last few years has made its appearance in daily news papers and television, and interactive visualization is common on the web. These are minimally interactive and often very simple visualizations. Can we develop ones that are engaging and easy to use? This is becoming critical as we are encountering the era of the democratization of data.

Finally, we are still developing visualizations and visualization systems based on experience, pragmatic research, and heuristics. We do not yet have a science of visualization. There have been attempts at automating the process: given data, automatically generate a visualization. It's clear that tasks need to be incorporated into the process. It's just as clear that evaluation and metrics can help solve this complex problem.

4. Issues of Evaluation:

In the early days of visualization research, rigorous evaluation was rarely performed; the assumption was that some visualization was better than no visualization for many tasks, and that if a new technique were developed, it was sufficient to simply place a couple of sets of images side by side and do a qualitative judgment of the results. More recently, there have been a large number of concerted efforts to incorporate a more formal evaluation process into visualization research, not only to enable quantification of improvements as they occur, but also to validate that visualization has measurable benefits to analysis and decision making. While many strategies have been developed and tested, there are many avenues of research toward improving the overall process.

The key point to make is that evaluation is a necessary component in moving a field from ad hoc methods to a real science, and we can learn from and build upon the evaluation strategies used in many other fields, such as human-computer interaction and sociology, to help validate the usefulness and effectiveness of techniques.

5. Issues of Hardware

Whenever computer technology advances, the applications that employ this technology must be reassessed to see how the advances can be leveraged. For visualization, there are several technologies that can and will have an impact.

(a) Hand-held displays.

Most people these days carry with them at least one form of digital display, whether it be mobile phones, PDAs, portable games, or tablets. While most visualization systems have been designed for desktop (or larger) displays, there are still significant opportunities to deliver interactive representations of information and data on these smaller devices.

Examples of potential applications abound. For aircraft, ships, and even buildings, having detailed presentations of wiring and plumbing diagrams, sensor output, and access paths can greatly simplify a technician's tasks. For crisis management during an emergency, police, firefighters, medical personnel, and other key players need interactive real-time access to information presented in a clear, unambiguous fashion. For those who monitor border crossings, rapid access to risk assessments, cargo manifests, and travel histories can help prevent entry by unwelcome individuals and material. The key is to develop visual solutions that make effective use of the limited display space and interactivity options.

(a) Display walls.

At the other extreme, large-scale displays, often involving multiple panels stretching 10–30 feet in each direction, are becoming more and more common, not only for control centers, but also for investigating large data and information spaces. While directly porting desktop solutions to these large displays is generally straightforward, it is not necessarily the most effective use of the display space. A better solution would be to redesign the visual analysis environment to arrange the displays of different types and different views of information in a way that supports the analysis. In this way, high-resolution displays can always be visible, rather than being covered in a typical desktop solution, thus requiring viewers to just move their head or shift their focus to see different views.

(b) Immersive environments.

Virtual and augmented reality systems have been frequently used within the visualization field. Early efforts, such as the virtual wind tunnel [52], enabled active, user-driven studies of the output of computational fluid dynamics experiments. Virtual walk-throughs and fly-throughs have been used in a diversity of fields, including architecture, medicine, and aeronautics. A key problem with this technology is the need to render the visualizations with minimal latency, which has spawned significant research in algorithm optimization. While the “killer application” for virtual environments has yet to be discovered, it will undoubtedly require significant visualization technology. Google glasses and similar devices providing an augmentation to the user's world are creating a number of opportunities for improved interactions and visualizations in dealing with the real world and the projected one.

(c) Graphical processing units.

The development of special-purpose graphics hardware has actually exceeded the growth in performance of general purpose CPUs, primarily driven by the computer game industry. There is much active research in the visualization community that focuses on harnessing this computational horsepower [249,399]. Due to the architecture of a typical GPU, existing algorithms designed for CPUs do not, in general, port directly to the GPU, but require a nearly total redesign. However, as more and more software and hardware engineers become versed in this programming paradigm, it is likely we will see a growing use of this

technology, not only for graphics, but also for complex algorithms in general.

(d) **Interaction devices.**

Each new device for user interaction with the computer opens up a wide range of possibilities for use in visualization. Voice/sound input and output have been extensively studied, though they are rarely an integral component of a visualization system. Tactile feedback is another area of great promise. Already, there have been efforts to train surgeons via virtual surgery, using force-feedback devices to mimic the feel of using probes and scalpels in surgical procedures [90]. Another avenue for development is to examine how different controllers used in modern game consoles could be employed to support visualization. The popular Wii input wand could be used for specifying actions via gestures. Other devices, such as head and eye trackers, may have significant potential in the visualization field. Interactive tables improve on the interactions with touch screens that have been used successfully for kiosks, and may lead to larger, less tiring environments. Brain control is another area providing great opportunities for controlling and interacting with visualizations. There are many opportunities for research on interaction metaphors.

6. Issues of Applications

Many advances in the field of visualization have been driven by the needs of a particular application domain. These advances are then often generalized to be applicable to many other domains and problems. Thus, research issues can be divided between those specific to a given domain and those involving the extension of techniques into other domains.

a) Depth-based innovations. To develop truly useful visualization tools for a particular domain, it is very important for the visualization designer to understand the domain and the tasks the domain specialist is trying to perform. Unfortunately, it is often the case that visualization designers just focus on the syntax or format of the data or information to be analyzed, which can easily result in tools that are either not useful/understandable to the domain analyst, or that provide little advantage over existing techniques. In a similar process, when domain experts attempt to design and develop visualization tools for their needs, they often produce tools with poor designs and user interfaces; this is mainly because they are generally unaware of the perceptual and cognitive issues involved in visualization design, as well as being unfamiliar with the range of different visualization techniques that have been developed for other purposes. In contrast, many of the most successful visualization systems have resulted from tight interactions between domain and visualization specialists, focusing on real problems that affect progress in a domain.

b) Breadth-based innovations.

Another direction of research is to broaden the number of applications in which data and information visualization can be applied. Indeed, it is hard to imagine an area in which visualization would not be applicable, as all areas of society are experiencing a glut of information, while at the same time, display devices have become ubiquitous. In many applications, visual information presentation is rapidly replacing much of the textual communication, such as weather reports, stock market behavior, health statistics, and so on. Daily schedules are often best captured in a graphical presentation of an hourly or daily calendar. Graphs are used to capture complex social networks, organizational charts, process flows, and communication patterns.

One key to successful research and development in the broadening of the visualization field is to find the appropriate visual representations and interaction metaphors for new domains and tasks. Another critical problem is the conversion of data and information

into a format that is amenable to existing visualization techniques. While many tools now accept input from standard database tables and spreadsheet files, much data and information is still stored either in proprietary formats or as unstructured files. Concerted efforts are needed, both in the visualization of unstructured data, as well as in automatic or semi-automatic conversion of unstructured data into structured forms, to tap into these rich sources of new visualization applications.